Net force first, then interpret the motion

LEARNING TARGET · R7 I can solve with $F = ma$ — always finding the NET force first.

SECTION A KEY VOCABULARY

Force sum	Add signed forces along the same axis.
Acceleration	Velocity change from net force per mass.
Velocity	Current motion; it can oppose a.
Speed trend	Opposite v and a: slows before reversal.

SECTION B KEY EQUATIONS + UNITS

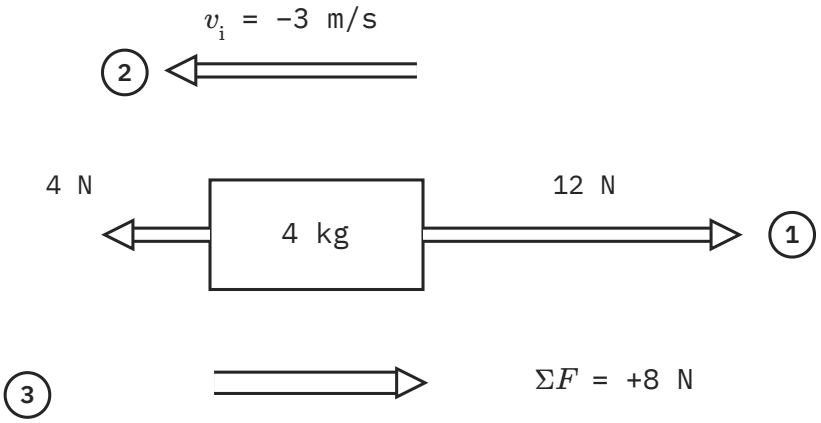
$$\Sigma F_x = ma_x$$

$$v_f = v_i + a_x t$$

ΣF_x	$F_{right} - F_{left}$
+	right

SECTION C COLOR IT · FORCE OPPOSES MOTION

The cart moves left.



after 1 s: v = −1 m/s; speed is smaller

- ☐ **1 · ORANGE** — Forces and separate net
- ☐ **2 · PURPLE** — Leftward velocity
- ☐ **3 · RED** — Slows for the first **1 s**

SECTION D WORKED EXAMPLE

A **4 kg** cart has forces of **12 N** right and **4 N** left. Its initial velocity is **−3 m/s**. **Find a and v after 1 s.**

GIVEN	EQUATION	WORK	ANSWER
m = **4 kg** F_{right} = **12 N**; F_{left} = **4 N** v_i = **−3 m/s**	Sum forces first. $\Sigma F_x = ma_x$. $v = v_i + a_x t$.	ΣF_x = 12 − 4 = **+8 N** a_x = 8/4 = **+2 m/s²** v = −3 + 2(1)	v = **−1 m/s** Still leftward; speed is now **1 m/s**.

SECTION E THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: FORCE TO MOTION

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

The force tells us how motion changes

Net force, acceleration, and a cart moving the other way

A force arrow pointing right does not guarantee that the object is moving right. It tells us about an interaction. The net force determines acceleration, and acceleration tells us how velocity changes. To describe the motion itself, we also need the velocity the object already has. Today we will follow that chain carefully.

Choose right as positive. Our **4 kg cart** has a **12 N force to the right** and a **4 N force to the left**. Both forces act on the cart, so they belong in the same horizontal sum: $\Sigma F_x = 12\ \text{N} - 4\ \text{N} = +8\ \text{N}$. Using only the larger force would ignore a real interaction. Adding both magnitudes without signs would ignore their opposite directions.

Use the force sum in **Section B** with right as positive: $\Sigma F_x = ma_x$. Thus $a_x = +8\ \text{N} \div 4\ \text{kg} = +2\ \text{m/s}^2$. The acceleration points right, matching the net force. The vertical forces are balanced, so they produce no vertical acceleration and do not add to the horizontal force sum.

Now include the starting velocity, **−3 m/s**. The cart initially moves left. With constant forces and mass, acceleration remains constant, so after **1 s** we use $v_f = v_i + a_x t$. Substitution gives $v_f = -3\ \text{m/s} + (2\ \text{m/s}^2)(1\ \text{s}) = -1\ \text{m/s}$. The cart still moves left, but its speed has fallen from 3 m/s to 1 m/s.

This is what positive acceleration means here: the signed velocity becomes more positive by 2 m/s each second. It does not instantly become positive. From −3 to −1 is an increase in velocity but a decrease in speed. I want you to say both parts when interpreting the answer, because the sign and the size describe different features of the motion.

If the same forces continue, the cart reaches zero velocity after $t = (0 - (-3))\ \text{m/s} \div (2\ \text{m/s}^2) = 1.5\ \text{s}$. It then moves right and speeds up. There is no need for the net force to reverse at the turning point. The same rightward acceleration first slows the leftward motion and then builds rightward motion.

In **Section C**, compare the cart's force and velocity arrows. The orange arrows identify forces and the purple arrow identifies initial velocity. The red label applies to the first second because speed decreases during that interval. The net-force arrow is a summary of the force sum, shown separately; it is not an additional force to add again. Keeping the picture and calculation connected helps prevent double-counting.

For our fictional asteroid, 2026-XJ, a force toward Earth tells us the direction of its acceleration due to that interaction. Its existing velocity might have components toward, away from, or across that direction. A useful forecast combines the net force with that initial motion and an elapsed time, rather than assuming the asteroid immediately travels along the force arrow.